KALAM (கலாம்) — AGRICULTURAL INSIGHTS ATLAS

Ambasamudram & Cheranmahadevi Taluks, Tirunelveli · Multi-Year EO Analytics (2023–2026)

CROPPING INTENSITY

0.2% Double-Cropped Cropland

AUDIT READINESS

0 Targeted Audit Parcels (GPS)

ANNUAL WATER DEMAND

40.16 Million m³ (MCM) Canal Requirement

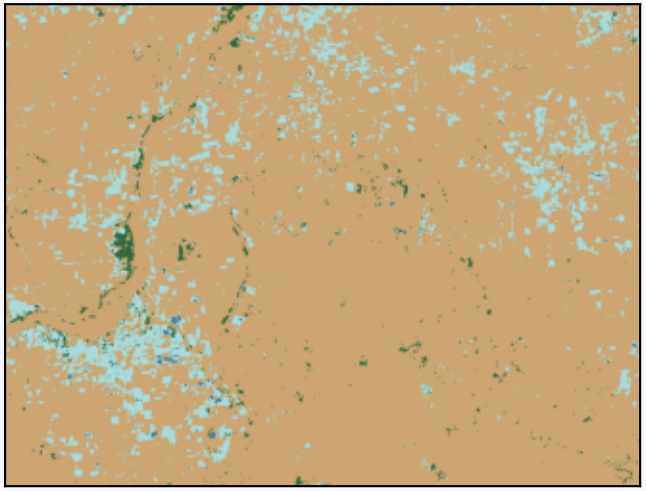
EXECUTIVE INSIGHT SUMMARY & POLICY ACTION ITEMS

- **Cropping Intensity:** 0.2% of pilot cropland is double-cropped (Kar & Pishanam paddy cycles), supported by perennial banana (1.4%) and single-crop systems (10.3%).
- **Transplanting Calendar:** Peak Kar transplanting occurs in Jul (18.6% of fields), while main Pishanam sowing peaks in Nov. 0.0% of Pishanam acreage experiences delayed transplanting beyond November.
- **Field Verification Priority:** 1.7% of parcels (34 fields) are flagged for targeted ground audit due to lower phenological confidence (<65%) or multi-year crop switching, enabling efficient sample-based field verification.
- **Village-Level Statistics:** Kallidaikurichi leads in paddy cultivation with 614.4 ha (68% of village cropland), while Kallidaikurichi features the highest banana concentration (7.7 ha, 1% share).
- **Irrigation Dependency & Water Access:** Banana shows sharp spatial decay from surface water channels: 1.4% of cropland within 250m is planted with banana, dropping to 0.3% beyond 2 km where paddy (77.4%) and rainfed crops dominate.
- **Crop Change & Farmland Dynamics:** 53.4% of paddy and 1.3% of banana parcels remained stable between 2023 and 2026. A net 120.3 ha (4.0%) transitioned from paddy to commercial banana, reflecting high-value crop diversification.
- **Crop Health & NDVI Anomaly:** 85.6% of cropland displays normal or above-normal vegetative vigor. Localized moisture or delayed transplanting stress is restricted to 14.4% (372.2 ha), primarily on tail-end irrigation distributaries.
- **Irrigation Water Demand:** Total estimated crop water requirement is 40.2 MCM/year. Dual-season paddy accounts for 87.8% (35.3 MCM) of agricultural water demand, while high-intensity perennial banana accounts for 2.0% (0.8 MCM).
- **Field Size & Land Fragmentation:** Extreme land fragmentation: 19.8% of agricultural parcels are smallholder plots (< 1.0 ha), covering 9.9% of total farmland. The mean parcel size is 1.64 ha (median: 1.41 ha).

Kalam © 2026 · Problem Statement 1 · SustainAI / Mapathon · Contains modified Copernicus Sentinel data

PHENOLOGY DYNAMICS & VERIFICATION PRIORITIZATION

01. Cropping Intensity

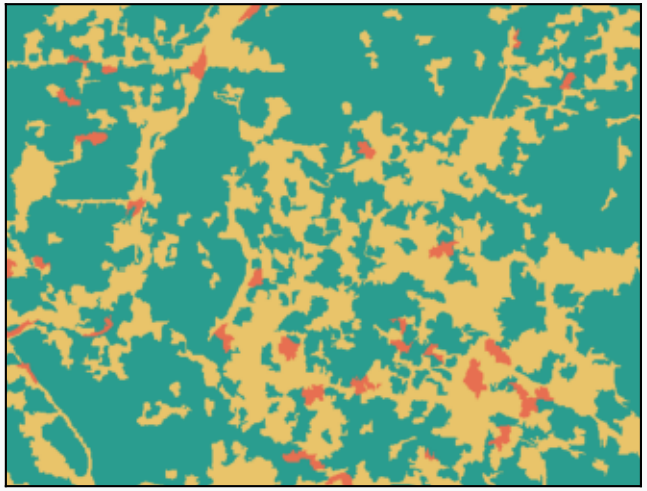


Fallow / Uncultivated | Single Crop | Double Crop | Triple Crop

02. Transplanting Calendar

Map Layer Ready

03. Field Audit Priority



Critical Priority | High Priority | Medium Priority | Low Priority

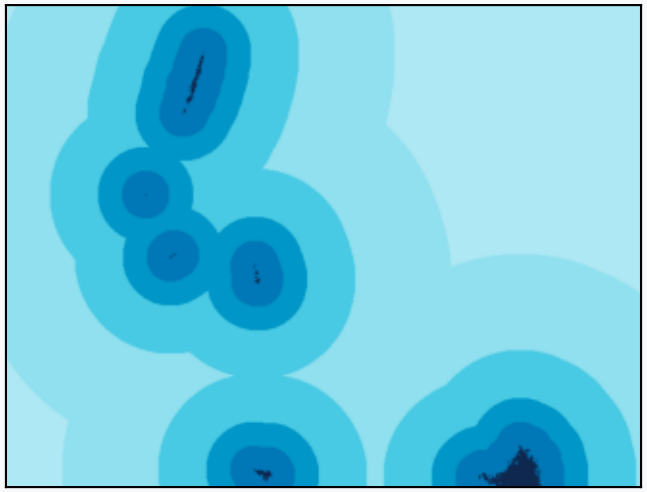
CANAL BUFFER DEPENDENCY & MULTI-YEAR CROP TRANSITIONS

04. Village Crop Allocations

Analytics Chart Ready

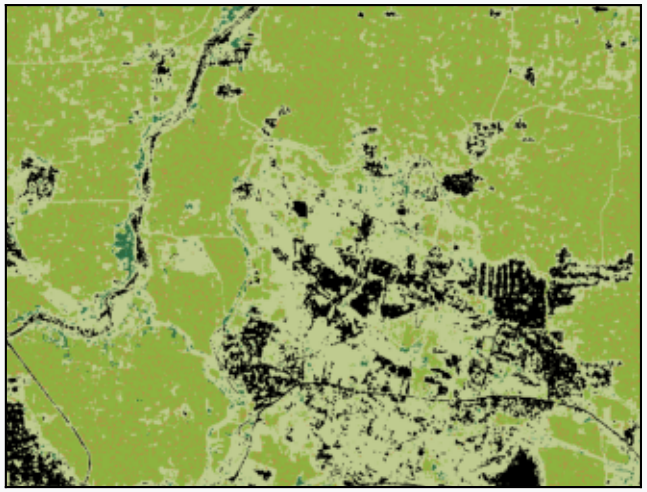
Paddy Area (ha) | Banana Area (ha)

05. Canal Buffer Decay



0 – 250 m | 250 – 500 m | 500 – 1,000 m | 1,000 – 2,000 m

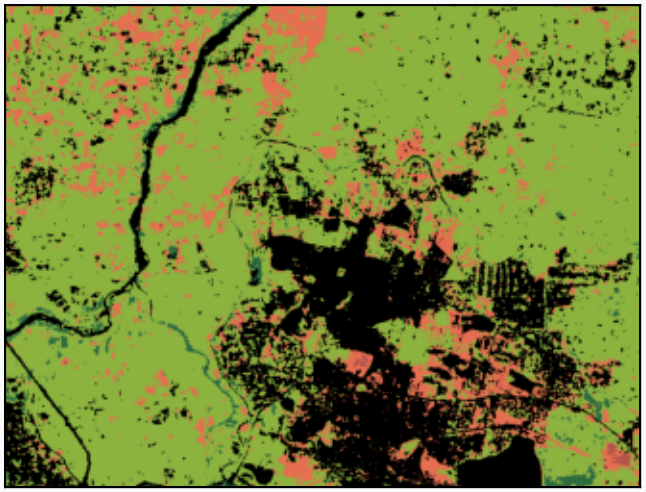
06. Crop Transitions (23–26)



Stable Paddy | Stable Banana | Paddy → Banana | Banana → Paddy

CROP VIGOUR, WATER DEMAND & PARCEL MORPHOLOGY

07. NDVI Vigour Anomaly



Severe Stress | Mild Stress | Normal Vigour | Above Normal

08. Water Demand Budget

Agronomic Layer Ready

Paddy (Kar) | Paddy (Pishanam) | Banana (Annual) | Other Crops

09. Smallholder Fragmentation

Agronomic Layer Ready

Marginal (< 0.5 ha) | Small (0.5 – 1.0 ha) | Semi-Medium (1.0 – 2.0 ha) | Medium (2.0 – 5.0 ha)